Chapter 18

Pharmacoeconomic Analyses and Modeling

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Learning Objectives:

Objective 18.1 Discuss the purpose of pharmacoeconomic analyses.
Objective 18.2 Explain pharmacoeconomic analysis approaches.
Objective 18.3 Describe common pharmacoeconomic modeling methods.

OBJECTIVE 18.1. DISCUSS THE PURPOSE OF PHARMACOECONOMIC ANALYSES

Pharmacoeconomics is a branch of health economics that generally focuses on the costs and benefits of drug therapy. All the pharmacoeconomic analyses estimate the cost of various intervention strategies, whereas most (but not all) also include health consequences. With any sort of cost analysis, perspective (cost to whom) matters. Perspectives for pharmacoeconomics analyses include the patient, provider, payer, and the broader societal perspective (including all costs) with the payer or societal perspective being the most common in the published literature.

Economic value may generally be defined as a "comparative analysis of the costs and consequences of a medical intervention or strategy as well as its next best alternative(s)," the value to whom is an important aspect of pharmacoeconomic analyses. Pharmacoeconomic analyses are increasingly used to assess the economic value of pharmaceutical products or medical devices.

The pharmaceutical industry invests enormous amounts of money in developing new health technologies.[1] Even though the success rate of developing newer health technologies is poor, this effort of selecting products that dramatically improve health and others do make incremental gains. After clinical trials, a pharmaceutical product must pass at least two additional regulatory steps. First, the marketing authorization (also called licensing) is granted, for example, in Europe it is the European Medicines Agency (EMA) while in the United States it is performed by the Food and Drug Administration (FDA). The EMA and FDA look at the quality, safety, and efficacy of drugs, by conducting a risk-benefit analysis. Once the pharmaceutical receives marketing authorization (i.e., the benefits outweigh the risks of the intervention), the pharmaceutical industry aims to set a price and achieves coverage and reimbursement. For example, this decision is made at the national level in Europe; however, in the United States, pharmaceutical companies can set a price for the approved drug in the market, but the drug reimbursement status needs to be granted by each health insurance program. In addition to benefit-risk considerations when a decision is made for reimbursement other factors must be taken into consideration such as convenience, affordability, and cost-effectiveness.

In a world with limited healthcare resources, cost containment is becoming important but innovating cheaper products is not common in healthcare. Almost all the time, newer technologies are costlier than the existing products but also provide added benefits over existing products. The questions to consider relate to whether or not the new technology is affordable and an efficient use of resources.

Pharmacoeconomic analyses vary mainly based on the types of outcomes used (Table 18.1). The analyses measuring and comparing the outcomes are considered to be full pharmacoeconomic evaluations.

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TABLE 18.1 Pharmacoeconomic Analysis Based on Types of Costs and Outcomes[1]

Analysis Type	Valuation of Costs[a]	Valuation of Health Outcomes	Calculation
Cost of Illness Analysis (COI)	$	None	At disease level
Budget-Impact Analysis (BIA)	$	None[b] or maximize various[c]	vs.
Cost-Comparison Analysis	$	None	vs.
Cost-Minimization Analysis (CMA)	$	Assume same	vs.
Cost-Consequence Analysis (CCA)	$	Natural units	vs.
Cost-Effectiveness Analysis (CEA)	$	Natural units	÷
Cost-Utility Analysis (CUA)	$	Utilities (e.g., quality-adjusted life years, QALYs)	÷
Cost-Benefit Analysis (CBA)	$	$	÷ or[d] −

[a] *Any currency.*
[b] *That is, determine the impact of an intervention/program on a designated nonfixed budget.*
[c] *That is, maximize some health outcome within a designated fixed ("capped") budget.*
[d] *Cost-benefit ratio (÷) or net of costs and benefits (−).*
Adapted from US National Library of Medicine, https://www.nlm.nih.gov/nichsr/hta101/ta10107.html.

Burden of Disease and Affordability

- Cost-of-illness analysis (COI): a determination of the economic impact of an illness or condition (typically on a given population, region, or country), e.g., of smoking, arthritis, or diabetes, including associated treatment costs.
- Budget-impact analysis (BIA): determines the impact of implementing or adopting a particular technology or technology-related policy on a designated budget, e.g., of a drug formulary or health plan.

Economic Evaluations Finding Value for Money Spent

- Cost-minimization analysis (CMA): a determination of the least costly among alternative interventions that are assumed to produce equivalent health outcomes.
- Cost-effectiveness analysis (CEA): a comparison of interventions regarding costs in monetary units and outcomes expressed in quantitative nonmonetary health units, e.g., reduced mortality or morbidity.
- Cost-utility analysis (CUA): a form of CEA that compares costs in monetary units with health outcomes regarding their utility and mortality, which is expressed in quality-adjusted life years (QALYs). QALY is a generic outcome measure that takes morbidity and mortality gains into account.
- Cost-consequence analysis (CCA): a form of CEA that presents costs and health outcomes in discrete categories, without aggregating or placing weights on the costs and health outcomes.
- Cost-benefit analysis (CBA): compares costs and health benefits (and risks), all of which are quantified in common monetary units.[1,2]

Pharmacoeconomic Evaluation Purpose

A COI study aims to measure the burden of a disease with regards to costs. This could be useful to prioritize between diseases but does not help in getting an efficient healthcare allocation for coverage and reimbursement decisions of a particular intervention. (A high-cost burden does not mean that treatments are available to reduce this burden.) As affordability is also important for short-run economic purposes, a BIA is often undertaken. By estimating the impact of the new drug on healthcare budgets, this method assesses the affordability of a healthcare intervention if the intervention is used within an environment as compared to not used within an environment.

A full economic evaluation aims to assess the value of money or efficiency of healthcare resources. The preferred type of economic evaluation is CUA because it allows the use of the same health outcome for all interventions and diseases, and thus to help decision makers to allocate resources efficiently.

Full Versus Partial Economic Evaluation

A full economic evaluation compares the costs and health outcomes of at least two interventions. Namely, an intervention of interest compared to its next best alternative(s). We distinguish five full economic evaluations: CBA, CCA, CEA, CUA, and CMA.

A partial economic analysis only evaluates one intervention in isolation or considers only one domain (costs or effectiveness). Under certain circumstances, COI and BIA are considered partial evaluations when they evaluate one disease or intervention in isolation.[3]

Levels of Application

Pharmacoeconomic analyses are useful in both national and local economic settings. In national or population-level decision making, such as the decision to provide access for and fund a drug, CEA and BIA are required components of the process. Other analyses are also used for specific purposes. At a national level related to research funding allocation decisions, COI could help prioritize between diseases. At a more local level, such as a hospital administrator, BIAs and CEAs are used for formulary inclusion decisions. Mini-health technology assessment (HTA) or hospital-based HTA is increasingly used to assess value in a hospital setting. The Danish Centre for Evaluation and Health Technology Assessment (DACEHTA) describes mini-HTA as a management and decision support tool based on reasoning involved in HTAs and may be used where a hospital is contemplating the introduction of new health technology.[4]

Thus, pharmacoeconomic analyses enhance public health decision making for both broader and also at local levels.[5]

OBJECTIVE 18.2. EXPLAIN PHARMACOECONOMIC ANALYSIS APPROACHES

Cost-of-Illness

Using the definition of the International Society of Pharmacoeconomics and Outcomes Research (ISPOR), a COI study or a cost-of-disease study aims to determine the total economic impact (cost) of a disease or health condition in society through the identification, measurement, and valuation of all direct and indirect costs. This form of study focuses on costs but does not address questions relating to treatment effectiveness or efficiency.[6]

COI studies generally measure medical costs; some COI studies also measure nonmedical costs. COI studies aim to assess the economic burden that a specific health problem or groups of health conditions impose on society regarding utilization of healthcare services and productivity losses. This research has the potential of informing policy-makers and decision-makers on the relative impact of diseases at the population level, assisting them in making projections of future healthcare costs and resource allocation decisions. Furthermore, COI studies aid in signaling to innovators the population-level unmet need regarding health losses and the potential for future interventions to reduce this unmet need.

The choice of cost methodology (i.e., top-down vs. bottom-up) for assessing both direct costs and losses in productivity varies depending on the availability of data.[7] For example, epidemiological data availability (i.e., disease prevalence, incidence, and associated mortality) varies by geography and other characteristics. Ideally, COI estimates should use a database linking epidemiological data, resource utilization, health services, and their actual costs (rather than the listed prices), alongside demographic data. Big data and patient registries are rarely available in Central and Eastern Europe, Western Asia and Africa (CEEWAA) countries. Nevertheless, even if these data sources were available, valuation of important cost components, such as patient out-of-pocket expenses and lost productivity of informal caregivers, would be rather difficult, and therefore might be excluded, thus underestimating the true COI.

The challenges on accuracy of methods used for assessing losses of productivity due to an illness are complex due to:

1. The human capital approach (HCA) and friction cost method (FCM) are different in approach. The HCA values the potential lost production because of an illness, whereas the FCM focuses on duration until the person is replaced by another one to continue to contribute to the productivity of the organization. Although appealing, using the FCM requires a tremendous amount of information that is difficulty to collect and subject to change with friction time, for example, on the unemployment rate in the country.
2. Should analyses include losses due to absenteeism or also due to presenteeism? Although the assessment of productivity losses resulting from absenteeism may be relatively uncontentious, this is not the case when presenteeism (i.e., the level of productivity while working) is concerned. These presenteeism-related costs are frequently omitted from COI studies.[7–9]

Budget-Impact Analysis

A BIA is an economic assessment that estimates the financial consequences of adopting a new intervention. It mainly deals with the affordability of new health technologies to a hospital or health system budget with limited resources.[10]

A BIA is usually performed in addition to CEA. A CEA analysis evaluates whether an intervention provides value (i.e., efficiency) relative to an existing intervention (with value defined as cost relative to health outcome). A BIA evaluates whether the high-value intervention is affordable.

A BIA is used to identify the total budget required to fund an intervention. BIA takes the true "unit" cost of an intervention and multiplies it by the number of people who utilize it. Consider the size of the population that utilize the intervention; if it is low, modeling might be required. Also consider if the intervention is replacing an existing standard of care (substitution) or being added to the existing standard of care (combination). There is also the possibility that the standard of care does not exist (e.g., due to patient intolerance of standard care). Modeling is used to fill gaps in data and it is subjected to sensitivity analyses to evaluate the impact of changes in the assumptions used in modeling.[11]

As a BIA is often used for resource allocation purposes, it takes a payer's perspective and uses a short-term time horizon (often 1−3 years). A BIA does not use discounting. Results should be presented on an annual or quarterly basis, or in whatever timeframe is relevant to the decision-maker.

The BIA focus is on the direct costs of specific resources needed to put the intervention into effect, such as supplies, equipment, and staff. Because the BIA uses a short-term time horizon, and overhead costs are fixed in the short term, these overhead costs are ordinarily excluded from BIAs. This distinguishes BIA from cost-effectiveness studies, which includes overhead costs. The overhead can account for a substantial part of the cost of operating a hospital or healthcare system and the CEA.[11]

Some differences between BIAs and CEAs are noted in Table 18.2.

Supply-side thresholds take resource allocation into account, e.g., estimates of the health foregone because an insurance company or other provider spends some of its available budgets on a new intervention and is therefore forced to reduce its funding of older interventions.[12] Budget impact is the difference between COI of new treatment and the existing standard treatment (Figs. 18.1 and 18.2).

TABLE 18.2 Comparison of Budget Impact Analysis With Cost-Effectiveness Analysis

	Budget-Impact Analysis	Cost-Effectiveness Analysis
Perspective	Payer	Societal
Time Horizon	Short term	Long term/Lifetime
Size of Population	Includes	Ignores
Model Inputs	Payer specific	Population average
Model Outputs	Cost	Cost and health outcomes
Discounting	No	Yes
Include Overhead Costs	No	Yes

Adopted from US Department of Veterans Affairs. Budget Impact Analysis. Health Economics Resource Center (HERC). https://www.herc.research.va.gov/include/page.asp?id=budget-impact-analysis.

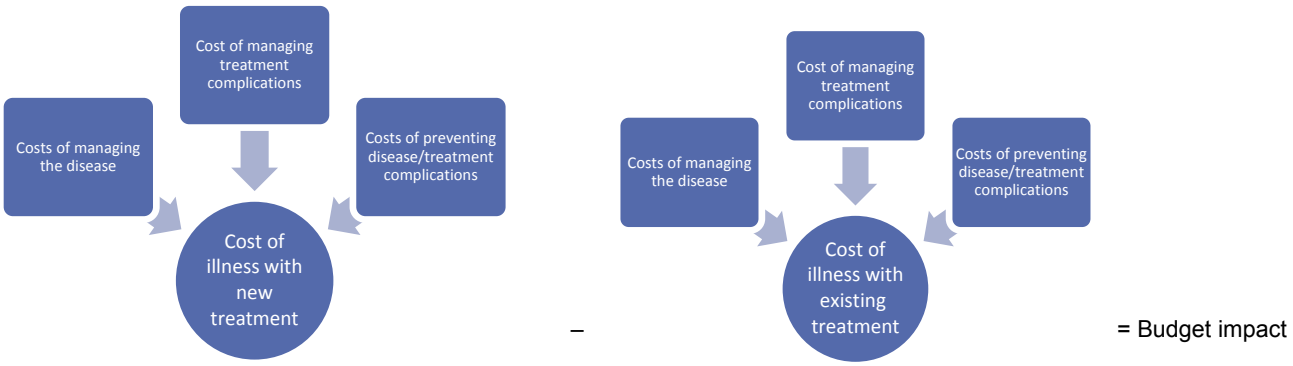


FIGURE 18.1 Framework of budget-impact analysis (difference in the costs of new and existing treatments).

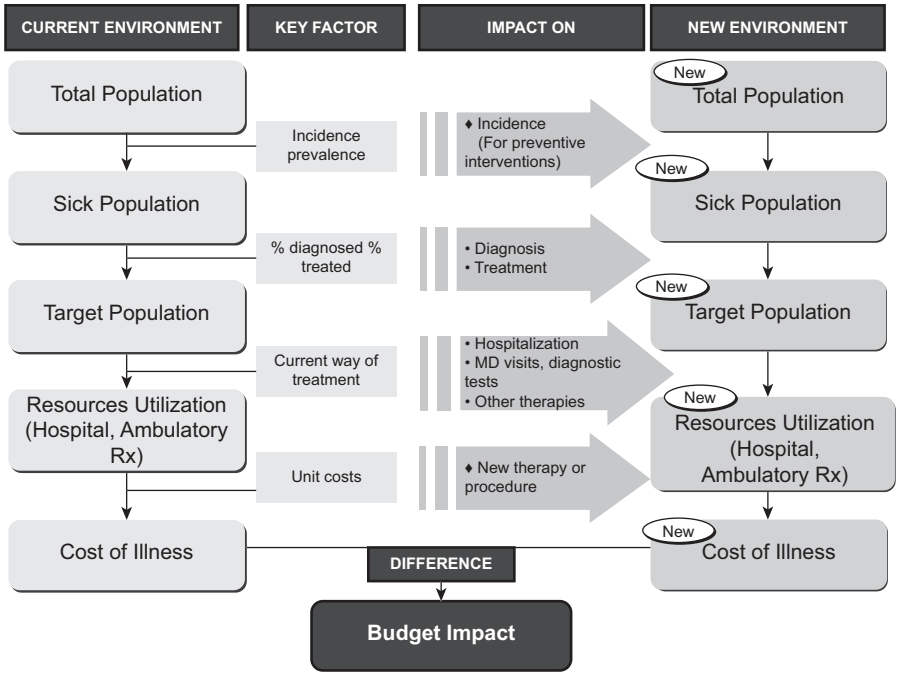


FIGURE 18.2 Budget-impact model. *Adapted from Brosa M, Gisbert R, Rodríguez JM, Soto J. Principios, métodos y aplicaciones del análisis del impacto presupuestario en el sector sanitario.* Pharmacoecon Span Res Artic. *2005;2(2):65–78. https://doi.org/10.1007/BF03320900.*

Cost-Comparison Analysis

As per the definition by ISPOR, a cost-comparison analysis compares only the costs associated with two or more alternative healthcare treatments or interventions. The process of identifying all the costs and their relative importance is often called cost-identification analysis.[6]

Cost-comparison analysis is not commonly used. When COI measures all relevant costs of the disease, including its treatment, cost-comparison analysis is only measuring the costs of different treatment options. Cost-comparison analysis compares the costs of two different treatment mixes (including all medical and nonmedical products and services). When the treatment options have multiple cycles and with complicated dosage regimens, cost-comparison analysis helps in totaling and comparing the differences in costs, e.g., costs of different chemotherapy regimens. In a complex treatment mix, it is essential to identify all the treatment costs, including drugs, devices, nutritional supplements, professional care costs, physiotherapy, acupuncture, etc. Policy-makers usually use COI instead of cost-comparison analysis.

Cost-Minimization Analysis

In the definition of ISPOR, a CMA is a type of pharmacoeconomic analysis comparing only two alternative therapies regarding costs because their health outcomes (effectiveness and safety) are found to be or are expected to be identical.[6]

The CMA considers the costs and health outcomes and compares two alternative therapies, but only in situations assuming the health outcomes are identical. In rare cases where the health outcomes are extremely similar, the decision-makers can prioritize the treatment option that is less expensive. CMA is commonly used to compare between the generic and branded versions of drugs with cost variation. As the generics are considered bioequivalent to the branded version of the same drug, the outcomes of therapy are expected to be identical. CMA could also be used to compare the use of generic and branded versions of drugs in different settings. The costs charged for the same drugs are different in a charity hospital compared to a corporate hospital. If we assume the treatment outcomes with similar drugs to be same in different settings, CMA could be performed. Therapy outcomes are the result of many factors involved in the care process, with drugs being one of the factors. The differences in professional care, nutrition, hygiene, genetic factors, etc. will alter the outcomes. Even the same drug will produce different outcomes depending on rational use approaches. The dose, timing of administration, dosage forms, interactions, etc. will influence the safety and effectiveness of drugs. CMA is criticized for

assuming identical outcomes when they are not similar. It is also doubtful that CMA is a full pharmacoeconomic analysis when the health outcomes are assumed to be equal but are not measured to test this equality assumption.[14]

CMA is useful in limited resource settings to decrease the cost when the outcome is the same for interventions of varying costs. Even in developed countries, healthcare budgets are limited. The policy is to provide services and products of acceptable quality, if not the best quality. Generic versions of drugs are preferred over costly brands. Price variation among different brands is significant for some products. Selecting a cheaper bioequivalent brand or generic is a common strategy to minimize costs.

Cost-Benefit Analysis

As per the definition of ISPOR, a CBA is an analytical technique derived from the economic theory that enumerates and compares the net costs of a healthcare intervention with the benefits that arise as a consequence of applying that intervention. For this technique, both the net costs and the benefits of the health intervention are expressed in monetary units.[6]

CBA is commonly used in many economics fields, including healthcare. Even though it is a common method of analysis in public projects, in clinical decision making, it is difficult to assign a monetary value to the clinical outcomes. CEA and CUA are used more commonly in healthcare. CBA is like risk-benefit analysis. Both could be described better as benefit-cost analysis and benefit-risk analysis because the expectation is that benefit outweighs costs. The value of the benefit is worth paying extra costs for the treatment, which is compared with a cheaper version. In CBA costs and benefits are represented in monetary units (e.g., dollar in the United States, pounds in the United Kingdom, etc.) As in CBA, the benefit (outcome) is in monetary units; the comparison is computational and straightforward. The costs and benefits of a healthcare intervention are compared with another standard intervention. This intervention could be a drug (most common), surgical procedure, pharmaceutical care service (e.g., counseling), or even a public health project (e.g., clean drinking water project and decreased expenses for infections averted):

$$\text{Incremental net monetary benefit} = (\text{Benefit of A} - \text{Benefit of B}) - (\text{Cost of A} - \text{Cost of B})$$

The result could be expressed as an incremental net monetary benefit of $10,000, for example. This example would suggest that intervention A is favored over intervention B as the benefits of A versus B less their costs, favor A. The interpretation of this example is that when we value health gains in monetary units, we expect to yield higher health gains in the monetary form of $10,000 for the average patient by opting for intervention A as compared to intervention B.

The difference between cost-benefit analysis and return on investment (ROI) should be understood. Although CBA helps in choosing a better treatment option or a project in comparison with an alternative, ROI measures the value of an investment. CBA is an evaluation and ROI is a valuation.

Return on investment: ROI is a performance measure used to evaluate the efficiency of a project or to compare the efficiency of different projects. ROI measures the amount of return on a project relative to its cost. To calculate ROI, the return (net benefit: benefit minus cost) of a project is divided by the cost of the project, with the result expressed as a percentage[15]:

$$\text{ROI} = (\text{Gain from investment} - \text{Cost of investment})/\text{Cost of investment} \times 100$$

Cost-Consequence Analysis

In the definition of ISPOR, a CCA compares the health intervention of interest to one or more relevant alternatives, listing the cost components and the various outcomes of each intervention separately. This type of economic analysis does not indicate the relative importance of the components listed and leaves it to the decision-maker to form his or her view.

The CCA is relatively difficult to perform as each outcome and its costs need to be separately assessed and evaluated. It is up to the decision-makers to consider the outcomes of their interest.

Cost-Effectiveness Analysis

As per definition by ISPOR, a CEA is a systematic method of comparing two or more alternative programs by measuring the costs and consequences of each. A distinguishing feature of CEA is that the consequences (health outcomes) of all the

programs to be compared must be measured in the same common units—natural units related to the clinical objective of the programs (e.g., symptom-free days gained, cases prevented, patients improved, life years gained). If there are just two alternative programs, their difference in cost (incremental cost) is compared to their difference in outcomes (incremental effect) by dividing the former by the latter. This ratio is known as the incremental cost-effectiveness ratio (ICER). If there are more than two alternatives, programs are compared on a systematic pairwise basis using their ICERs.

Many believe that the other full pharmacoeconomic analyses like CBA, CUA, CMA, CCA, etc. are subtypes of CEA. Technically, CEA has differences to other analyses in the denominator of effectiveness (outcome) when calculated. CEA uses natural units of outcomes such as life years gained, symptom-free period, blood pressure, etc.:

$$\text{Incremental cost effectiveness ratio (ICER)} = \frac{\text{Cost}_A - \text{Cost}_B}{\text{Effect}_A - \text{Effect}_B}$$

The result could be written as $45,000 per life year saved. The interpretation of this example is that for every $45,000 invested in intervention A to treat the stated population, we would expect to observe one additional health unit gained (life year in this example) as compared to treating the population with intervention B.

Cost-effectiveness of new health technology (innovative drug) in comparison with an existing health technology (comparator standard drug) could be explained using a cost-effectiveness plane. Effectiveness is on the horizontal axis and cost is on the vertical axis. The "X" in the center represents the cost and effectiveness of the existing health technology. Once the cost and effectiveness of the new technology are measured, they can be spotted anywhere in the four quadrants that help in decision making (Fig. 18.3). Each quadrant is numbered to make things clearer.

First quadrant: If the cost and effectiveness of the new health technology fall within the first quadrant this indicates that the new drug is costlier and less effective compared with the standard drug at X. The new drug shows "strong dominance" in being expensive and with poor outcomes. The fourth quadrant also shows strong dominance. The principle of "strong dominance" leads to easy decision-making: accept or reject.[16] In the case of the first quadrant, it is to reject. There is no sense in paying extra for a new intervention that produces poor outcomes compared to existing standard treatment.

Second quadrant: If the cost and effectiveness of the new health technology is spotted in the second quadrant, a CEA needs to be performed. Here, the strong dominance principle is not used as the new therapy because it is clearly not the standard therapy. The downside is increased cost. Now the question is whether the additional effectiveness is worth paying the additional cost. Is the new therapy value for money spent? Well, the first thing to clarify is that the information from the cost-effectiveness plane or ICER may or may not be used in decision making. This CEA provides only some information, not the complete information required in decision making. Many factors like cultural, political, trade, budgetary, and other administrative factors will influence decision making. CEA is mostly a reasonable attempt to make pharmacoeconomic information clear. The usually used strategy is to draw a line through quadrants 3 and 2. This is called the cost-effectiveness threshold. It is a limit above which the decision will be to reject and below which decision will be

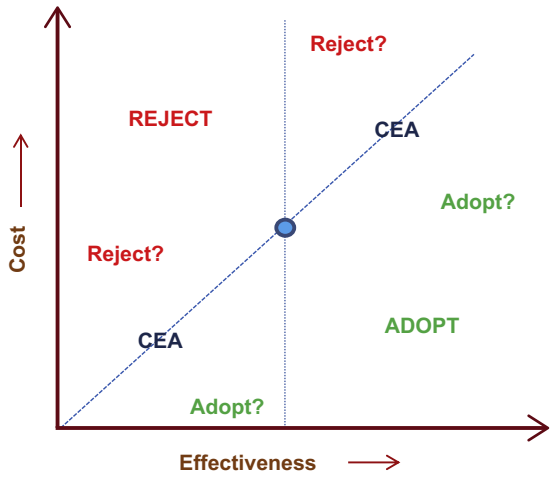


FIGURE 18.3 Cost-effectiveness plane in decision making. *Adapted from National Information Center on Health Services Research and Healthcare Technology (NICHSR). HTA 101: V. Economic analysis methods. NIH.*

to accept. This threshold is selected based on many parameters that affect affordability and expected gains in health. The concept of opportunity cost is really important. A trade-off should be made considering the health gain due to other interventions with the money saved by not buying this new costly drug, with the loss in health if this new drug is not used.[12] Consider the example of a new treatment that cures cancer compared with an existing treatment that could only mitigate the condition. Surely the new treatment will increase life years gained. Because recovering to near normal life is "invaluable," the cost-effectiveness threshold could be higher. A society's cost-effectiveness threshold—which indicates its willingness to pay (WTP) for improvements in health—can also be inferred from its budget for healthcare expenditures. The interventions are to be ranked from lowest incremental cost-effectiveness ratio (ICER) to the highest. Lower ICERs are preferred over higher. Going down in the ranking, at certain points all the available resources will be exhausted and further technologies could not be selected. The cost-effectiveness of the last intervention in the list might be costly and least preferred but comes within the available resource is the society's WTP threshold. It is the highest price the society is willing to pay for the least preferred technology due to higher expenses per quality-adjusted life year (QALY) gain. Healthcare interventions need to be cost-effective and appropriately priced.[17]

Third quadrant: In developing economies with a significant poor class in society, pharmaceutical innovation could focus on to innovate cheaper treatment options often compromising the outcomes. This, of course, creates ambiguity. Decision-makers might need to consider certain thresholds to assist in decision making. Is it still an acceptable level of outcomes for the money spent? The trade-off is what other significant health outcomes could be achieved with this saved money. Decision making becomes even tougher on ethical grounds for accepting new interventions that are worse than existing interventions. The society's cost-effectiveness threshold is called the willingness to accept (WTA) for having fewer outcomes. WTA often exceeds WTP.[18]

Fourth quadrant: Is a situation of reduced cost with increased effectiveness. It is a strong dominant to decide on acceptance of the intervention. The question here is: is it is possible? Such a situation rarely happens in healthcare. As clinical trials are costly and the success rates of new healthcare technology are poor, new intervention is almost always costlier compared to an older intervention.[19] However, this can be possible when we prevent additional costs due to the positive effects of the new treatments.

Cost-Utility Analysis

In the definition of ISPOR, a CUA is a methodology of economic analysis that compares two or more alternative choices regarding both their costs and their outcomes, where the outcomes are measured in units of utility or preference. The purpose of the analysis is to compare, using cost-effectiveness, two or more alternative choices regarding being both clinically meaningful and compared with other economic analyses. CUA can be considered as the "gold standard" methodology for evaluating the cost-effectiveness of healthcare choices.[4]

CUA is considered as a type of CEA and is sometimes called CEA. The only technical difference is that CUA uses QALYs as the denominator in calculating the ICER. Being a universal unit of health outcome, QALYs have been used often. For example, as per NICE, in the United Kingdom £25,000 or up to £30,000 (with conditions) per QALY gained by a new intervention is an acceptable value for money spent to purchase the new intervention. This limit is called the cost-effectiveness threshold. The cost-effectiveness threshold is defined as the "maximum value of money per health outcome that a jurisdiction decides to pay for adopting a technology or an intervention."[20] QALY even though common it is also criticized for its limitations. QALYs gained in geriatrics are not the same as QALYs gained in pediatrics, neither does it differentiate between QALYs gained by a mother of four young children compared with a similarly aged woman with no dependents.

A QALY is a universal health outcome measure applicable to all individuals and all diseases, thereby enabling comparisons across diseases and across programs. QALY is a measure of health benefits of medical interventions, whereas disability-adjusted life years are a measure of the burden of diseases. QALY measures both mortality and morbidity benefits with medical interventions. QALY is the sum of the number of years lived after the intervention with a certain quality of life (regarding utility). Utilities are measured on a scale of zero (representing death) to one (representing perfect health). Some disease conditions are considered worse than death, thus negative utility value could be considered.

The utility is a quantitative expression of an individual's *preference* for, or desirability of, a particular state of health under conditions of uncertainty. Direct elicitation methods to calculate utility are standard gamble (SG), time trade-off (TTO), or visual analog scales. Health-related quality of life measures are not considered as utility measures, e.g., SF-36 and disease-specific instruments. Indirect elicitation methods include, e.g. EQ-5D and SF-6D.

Standard gamble: In the SG method, patients are asked how much of a risk of death they are willing to incur to be cured of the condition.It is a hypothetical choice of taking risks with treatment that has the potential to cause death as an adverse

event or cure the disease. A chronic disease state with arthritis and a nonfunctional knee could make the patient choose to undergo knee replacement surgery. Hypothetically, the patient preference is to ask if a probability (p) of 80% is the chance of returning to normal health after surgery (1−p is a 20% chance of immediate death). If the patient accepts to undergo the surgery, the probability of returning to normal health is decreased until a point, e.g., 70% (1−p, the chance of dying is 30%), where the patient is unable to decide yes or no to undergo surgery. In this case, the utility is 0.7. If another patient is *does not* accept to undergo surgery with an 80% chance of returning to normal health, the probability of returning to normal health is increased, e.g., 85% (1−p is 15%), where patient is unable to decide yes or no to undergo surgery. In this case, the utility is 0.85. The point of indifference (not able to decide to undergo surgery or remain in the disease state) is the utility level. The utility (preference) scores for severe cases will be lower compared with not so severe cases.

Time trade-off: In the TTO method patients are asked to rate how much healthy life they are willing to give up to be cured of the condition. It is a hypothetical choice between living with less health for a specific amount of time versus living in perfect health for a shorter period of time. Here the choices are given to a patient to continue staying in ill health for the period of life expectancy or undergo a treatment to be fully healthy for a shorter duration of time followed by death. With the suggested new treatment (for full health), if the reduced life expectancy is 30 years compared with 50 years of continued chronic disease, depending on the patient's preference to accept or reject the new treatment, changes are made. If the patient accepts, the number of years they are expected to live with full health followed by death is *reduced* until the patient is not able to choose yes or no to the treatment. On the other hand, if the patient is not accepting 30 years in full health as an alternative to living 50 years with a chronic disease, the number of years expected to live with full health followed by death is *increased* until the patient is not able to choose yes or no to the treatment. If the number of years where the patient is unable to choose yes or no to the treatment is 35 years, 35/50 = 0.7 is the utility.[16]

Visual analog scales: In visual analog scales, patients are asked to select a score for their present health status. These are patient estimated preference scores for different disease states. In the visual analog scale of 0 (death) to 100 (perfect health), if a patient scores 50, that indicates a utility score of 0.5:

$$\text{Cost-utility ratio} = \frac{\text{Cost}_A - \text{Cost}_B}{\text{QALY}_A - \text{QALY}_B}$$

One year lived in full health (1 utility) is 1 QALY. One year lived with half health (0.5 utility) is 0.5 QALYs. A half year lived with half health (0.5 utility) is 0.25 QALYs.

The result of a CUA could be written as, e.g., $50,000 per QALY gained. The interpretation of this example is that for every $50,000 invested in intervention A to treat the stated population, we would expect to observe 1 additional perfect year of health gained as compared to treating the population with intervention B. Using thresholds of maximum payment per QALY gained is commonly used to ease the decision-making process considering affordability.

Perspectives in Pharmacoeconomic Analysis

Based on differences in the perspectives selected, identification, measurement, and comparison of the costs and consequences change in pharmacoeconomics. The most comprehensive perspective is societal. It includes the perspectives of all stakeholders in healthcare and is thus complex and expensive to consider. The US FDA mandates that any pharmacoeconomic analyses submitted to the regulators should also include a societal perspective. This is called a reference case.[19] When the health insurance companies and employer perform pharmacoeconomic analyses, the payer perspective is used. When studies focus on the services of healthcare providers like hospitals and healthcare professionals, a provider perspective is used. Finally, the important patient perspective covers the viewpoints of the patient. Common perspectives used in pharmacoeconomic studies are payer and societal.[2]

The merits and demerits of pharmacoeconomic analyses approaches are listed in Table 18.3.

Sensitivity Analysis in the Pharmacoeconomic Analysis

Some level of uncertainty is involved in estimations of costs, outcomes, and other variables used in a cost analysis. Certain variables that influence significant changes in the results should be estimated. To address uncertainty, sensitivity analysis should be performed. A sensitivity analysis may find, e.g., that including variables like indirect costs, or assuming the use of generic as opposed to brand-name drugs in a medical therapy, or using a reasonable higher discount rate in an analysis, changes the cost-effectiveness of one intervention compared to another.[1]

TABLE 18.3 Merits and Demerits of Pharmacoeconomic Analyses Approaches

Merits	Demerits
Cost-of-Illness (COI)	
• Easy to calculate	• Not comparing outcomes with costs. • Wary from settings to settings. • Not a universal measure, results vary in different settings.
Cost-Comparison Analysis	
• Easy to calculate	• Not comparing outcomes with costs. • Not a universal measure, results vary in different settings.
Cost-Minimization Analysis (CMA)	
• More useful in resource-limited settings.	• Assumes the outcomes are similar, may not always be true. • Cost-effectiveness analysis is a better alternative even if the outcomes are similar. • Not a universal measure, results vary in different settings.
Budget-Impact Analysis (BIA)	
• Critical analysis dealing with affordability of products and services within a budget.	• Outcomes not measured. • Without cost-effectiveness analysis, BIA has less value in decision-making. • Not a universal measure, results vary in different settings.
Cost-Benefit Analysis (CBA)	
• Compares both costs and outcomes.	• Monetary units for outcomes is not natural, and sometimes unethical. • Not a universal measure, results vary in different settings.
Cost-Consequence Analysis	
• Compares both costs and discrete outcomes. • Decision-makers could find in-depth analysis of each measurable outcome (consequences).	• Difficult to perform (calculate). • Not easy to summarize by the decision-makers. • Not a universal measure, results vary in different settings.
Cost-Effectiveness Analysis (CEA)	
• Compares both cost and outcomes. • Outcomes are measured in natural units. • Combines multiple outcomes.	• Not a universal measure, results vary in different settings. • Does not consider equity considerations.
Cost-Utility Analysis (CUA)	
• Universal unit, QALY. • Gold standard method. • Combines multiple outcomes.	• QALYs may not have the same value in people with different socioeconomic statuses.

Discounting in Pharmacoeconomic Evaluations

Discounting of costs and outcomes is performed if the costs and effectiveness outcomes are considered beyond 12-month time periods. The present value of money, as well as better health, is higher than future costs and outcomes.

OBJECTIVE 18.3. DESCRIBE COMMON PHARMACOECONOMIC MODELING METHODS

The ISPOR book of terms definition is: Modeling refers to the development of a simplified representation of a system (e.g., a population). A particular model may be analytical, visual, or both. In pharmacoeconomics specifically or health economics in general, analytical models can be used to pose and answer questions about interventions that cannot be directly answered by clinical trials due to time and financial constraints. Decision trees are both analytical and visual.[21]

Modeling is especially required when data from research observations are not available. Modeling is not a substitue for generating research data. There are possibilities of it going wrong.

In many situations, the data are insufficient for pharmacoeconomic analysis and related studies to make more informed decision making. In such cases, analytical, diagrammatic models, or mixed models are used to generate clearer information.

Analytical models are mathematical models that use data available from cost-effectiveness studies to prepare further interpretations of the data. Decision trees, the Markov models, and simulation models are commonly used to improve the quality of information in decision making.[19] As the models are based on certain assumptions, there is the possibility that these assumptions may be wrong. Pharmacoeconomic analyses along with modeling supplies information that helps its decision making. This information may be partly considered or fully rejected by decision-makers considering the limitations of such tools in a given context. Thus, these methods are just tools to assist in decision making, and it is not mandatory that the decision should be as per the results of such studies.

Decision Analyses

Decision analyses can be operationalized through decision trees or simulation models. Discrete event simulation models are useful in emergency care or in the transmission of infections. The events must be mutually exclusive. Individuals progress through the model only if they experience a new event (e.g., moving from the ED waiting room to an ED exam room).[22]

State-transition modeling is an intuitive, flexible, and transparent approach to computer-based decision-analytic modeling. A state-transition model can be used to model a cohort of patients, in which case it is called a Markov cohort model. A state-transition model can also be used to model individuals; in this case, it is called a microsimulation (first-order Monte Carlo) model. Most frequently, state-transition models are used in the evaluation of risk factor interventions, screening, diagnostic procedures, treatment strategies, and disease management programs. Discrete event simulation is another rarely used method with flexible frameworks used especially in constrained resources. Because it is used to represent complex systems, it is difficult to develop, implement, and analyze.[23,24]

Regardless of their structural form, there are certain similarities across all healthcare decision analyses:

- all decision analyses require the clinical- and policy-relevant features of the problem, the time frame of the analysis, and the relevant patient population to be specified.
- they all require information on the probability of experiencing a health state or a health event. In decision analysis, these are called "transition probabilities."
- all healthcare decision analyses require information on the payoff(s) associated with a health state or health event. The payoff(s) could be the cost, the health effect, or both.
- due to the limitations of data availability, almost all healthcare decision analyses use inputs from multiple studies (e.g., results from one study evaluating the efficacy of an intervention and estimates of the costs of treating the disease from another study, etc.).[22]

Elements of a Tree (Fig. 18.4)

- There is one decision node at the root.
- The branches of the initial decision tree represent all the strategies that are to be compared.
- There is a series of chance nodes off every strategy branch.
- The outcomes are depicted at the end of each pathway.

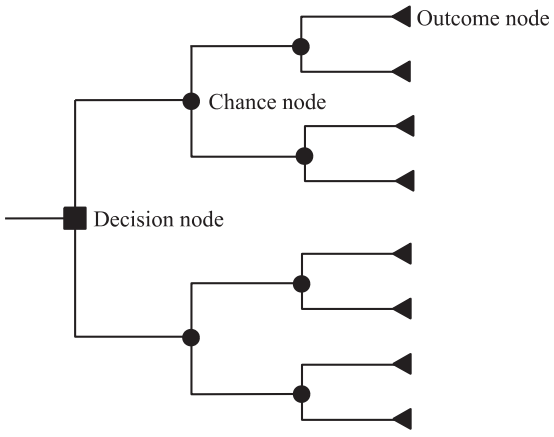


FIGURE 18.4 Decision tree.

Markov Model

Markov models represent disease processes that evolve over time and are suited to model progression of chronic disease; this type of model can handle disease recurrence and estimate long-term costs and life years gained/QALYs.

A Markov model is structured around health states and movements between them. Hence, the natural history of the condition becomes very important in designing the model. Within a model, health is split into distinct categories (or health states) that must be mutually exclusive and cover all the people in the model (i.e., all people must fit into a health state at any given point in time). An individual can only be in one state at a time and will remain in that state for a specified or fixed period of time, i.e., a cycle. At the end of each cycle, the individual/patient can either stay in the same health state (i.e., cycle) or move to another health state.

Let us look at an example. Imagine a simple model in which we have three health states: well, disease, and dead. This model can be shown in the form of a picture using an influence diagram to show the health states and movements between them (Fig. 18.5). If we look into the figure, we notice that three health states are mutually exclusive, i.e., patients can only be well, have the disease, or be dead within the model. Individuals in the model move at the end of each period or cycle along an arrow. For example, if people are in a well healthy state, then they can move to the disease state or the dead state or can be dead within the model. Similarly, people can move from the disease state to the well state or to the dead state or remain in the disease state. Once people are in the dead state, they cannot move from that state to any other state. However, while thinking about the patient journey, one needs to think about how long will people stay in each state, how often will people stay in each state, and the associated costs and outcomes for each state. One way to decide how long each period or cycle is is to consider the "cycle length," i.e., the minimum time people spend in a state (all members of the cohort will spend at least one cycle in the state they begin with). The cycle length is influenced by different factors and is often informed by the smallest clinically meaningful item included as a distinct event. Usually, the cycle should be short enough to consider the changes of clinical effects and resource use in patients between the cycles.

The elements of Markov models are:

- Mutually exclusive and exhaustive.
- The specific cycle length refers to a fixed period of time.
- Transition probabilities depict the transition from one state to another at the end of a single cycle.
- Markov states represent the cost and utility of spending one cycle in that state.

Steps in conducting a Markov model are:

1. Define states and allowable transitions
2. Choose a cycle length
3. Specify a set of transition probabilities between states
4. Assign a cost and utility to each health state
5. Identify the initial distribution of the population
6. Methods of evaluation

The merits and demerits of a decision tree and Markov model are listed in Table 18.4.

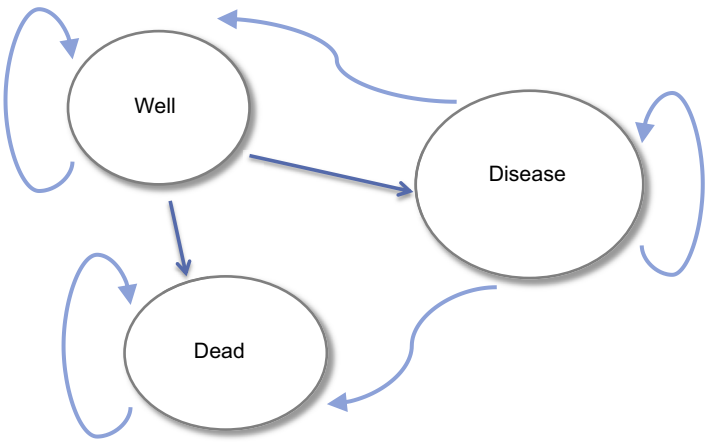


FIGURE 18.5 Markov model.

TABLE 18.4 Merits and Demerits of Pharmacoeconomic Modeling

Merits	Demerits
Decision Tree	
• Used for "one-off" decisions. • Suited for acute care, once-only diseases, short-term diagnostic/screening decisions.	• Less suitable for longer-term outcomes, though possible to add branches, but these are not efficient. • Difficult to handle disease recurrence. • Need to be able to assess full implications of each possibility of the patient pathway.
Markov Modeling	
• Can represent disease processes that evolve over time. • Suited to model the progression of chronic disease. • Can handle disease recurrence. • Estimates long-term costs and life years gained/QALYs.	• No account is taken of history. • Assumes uniform populations are an equal and constant risk. • May overcome these limitations by using a larger number of states. • Alternatively use other methods (individual sampling models, discrete event simulation.)

CONCLUSION

Pharmacoeconomic analyses are useful in assessing the value of pharmaceutical interventions and their affordability. The choice of the method and perspective depends on the purpose of the study. A comparison of merit and demerits helps in differentiating between pharmacoeconomic analyses. Pharmacoeconomic analyses include COI, BIA, cost-comparison analysis, CMA, CEA, CUA, CCA, and CBA. Modeling supports the generation of pharmacoeconomic data.